Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building production **Python**, **FastAPI**, **SQL**, and **CI/CD** systems for data-intensive cloud applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a multi-tenant platform on **Azure SQL** and Blob Storage with row-level isolation and schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built **FastAPI** data integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed production **Python** and Flask services with React interfaces for uploads, grading history, and feedback, reducing turnaround 60% while supporting 150+ instructors and 10K+ daily requests reliably.
- Implemented concurrent batch processing and PDF, DOCX, and ZIP ingestion through **REST APIs** and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery and usage tracking.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer for offline reinforcement learning, improving long-horizon trajectory planning 18% in simulation while developing Python pipelines that processed 100K+ autonomous-driving trajectory logs reproducibly.
- Designed a spatial-temporal encoder combining HD-map geometry and ego-agent trajectories through multi-head attention, improving closed-loop planning stability 15% while evaluating alternatives in reproducible simulation experiments against baselines.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Troubleshot and optimized large-scale API integrations using object-oriented patterns, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and maintainability while reducing technical debt 20%.
- Automated nightly builds, installations, **PyTest unit tests**, integration tests, and reports through Jenkins across Windows, Linux, and macOS, collaborating with Microsoft and GitHub stakeholders throughout Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight modular microservices for authentication, resources, customer records, billing, leads, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSocket updates across business workflows reliably.
- Automated **Docker** builds, unit, integration, and end-to-end tests, and staged deployments through Jenkins CI/CD, achieving 95%+ validation while using Sentry, Azure Monitor, and Kibana for production troubleshooting.

Skills

Languages & APIs: Python, SQL, Java, TypeScript, JavaScript, FastAPI, Flask, Pydantic, REST APIs
Architecture & Data: OOP, Data Structures, Algorithms, API Design, PostgreSQL, SQL Server, Azure SQL, SQLite, Snowflake
Cloud & DevOps: Microsoft Azure, AWS, S3, Docker, Git, GitHub, GitHub Actions, CI/CD, Jenkins, Linux
Quality: Unit Testing, Integration Testing, Regression Testing, PyTest, Code Reviews, Debugging, Refactoring, Agile/Scrum, Technical Documentation, Runbooks, Monitoring, Cross-functional Collaboration

Projects

FactorTrail | *Python, FastAPI, Pydantic, SQL, Pandas, PyTest, GitHub Actions*

- Built a **Python/FastAPI** factor-research service with Pydantic models, SQL audit trails, and point-in-time availability controls; processed 24,016 observations with zero look-ahead violations across 36 passing tests.

ClauseTrace | *Python, FastAPI, Pydantic, PyTest, BM25, TF-IDF*

- Built a Python/FastAPI contract retrieval service with **Pydantic** models, pytest, and matter-level access controls; improved hit@1 38% to 0.55 while recording zero citation mismatches and cross-matter leaks.

Chat2SQL Code | *Java, TypeScript, Snowflake, PostgreSQL, Docker*

- Built a Java and TypeScript database workspace supporting 10+ systems, including **Snowflake** and PostgreSQL, accelerating query authoring, schema exploration, and data management through reusable cross-platform service components.